

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Term Project Presentation Requirements

Purpose

The presentation should communicate the biological question, modeling approach, primary results, and ecological interpretation of your term project. Focus on the decisions and findings that the audience needs to understand. You do not need to present every processing step or model output. Projects with weak, unexpected, or conditional results can still produce strong presentations when the limitations are explained clearly.

Presentation format

- Presentations will take place Thursday, December 10 during the 1:00 to 3:00 PM final examination period, unless otherwise arranged.
- Plan for approximately 10 to 12 minutes of presentation.
- Allow approximately 3 minutes for questions.
- Aim for approximately 8 to 12 content slides, plus references.
- Submit your slides as a PDF through Canvas.

Remain within the time limit. Practice the presentation in advance and identify material that can be omitted if time becomes limited.

Required content

1. Research question and biological motivation

Introduce:

- The study organism or ecological system
- The geographic context
- The research question
- Why the question matters ecologically or for conservation
- Your biological expectations or hypotheses

Use relevant ecological literature to establish the motivation.

2. Occurrence data and study region

Briefly explain:

- Occurrence-data source
- Major filtering decisions
- Final number of records

- Geographic and temporal coverage
- Important sampling limitations

Include a clear occurrence map.

3. Accessible area and environmental data

Show and explain your definition of M. Identify:

- The biological reasoning behind M
- How background locations were sampled
- Environmental predictors
- The biological reason for including the predictors
- Important differences in spatial resolution or time period

4. Modeling approach

Explain:

- The primary model or models
- How model complexity was controlled
- How data were divided for fitting and evaluation
- The evaluation measures used
- The major assumptions tested

Focus on why you made these choices. Avoid presenting lengthy code or every software setting.

5. Results

Present the results that directly address the research question. Include:

- Model-evaluation results
- A clearly labeled relative-suitability map or other primary output
- Relevant response curves or variable-importance results
- At least one comparison showing robustness or sensitivity

Explain legends, colors, axes, and units. Maps should identify the study region and occurrence locations when relevant.

6. Interpretation and uncertainty

Explain:

- The main biological interpretation
- Whether the result matched your expectations
- Whether the interpretation was stable, conditional, or not robust
- Important limitations or alternative explanations
- What the model does and does not estimate

7. Conclusion

End with a concise answer to the research question. Identify the most important ecological conclusion and one useful next step.

Slide and delivery guidelines

- Use readable text and labels.
- Keep text brief and explain details verbally.
- Use figures and maps as evidence.
- Avoid screenshots of code or large tables of model settings.
- Define technical terms.
- Prepare to answer questions about M, predictor choice, model evaluation, and uncertainty.